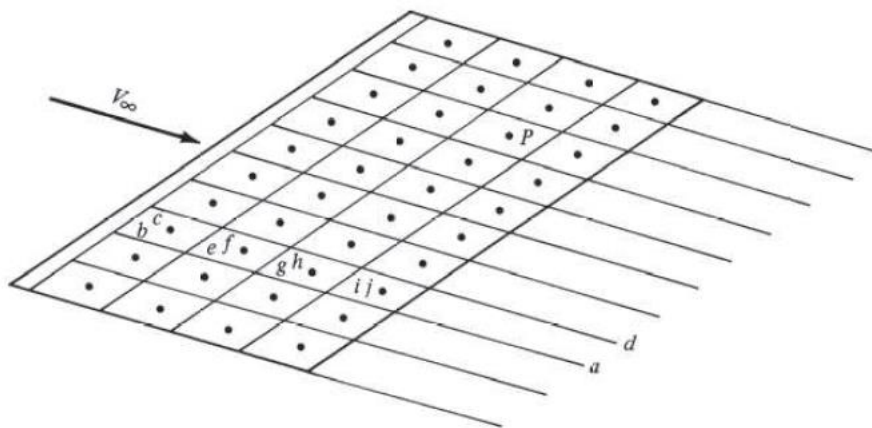
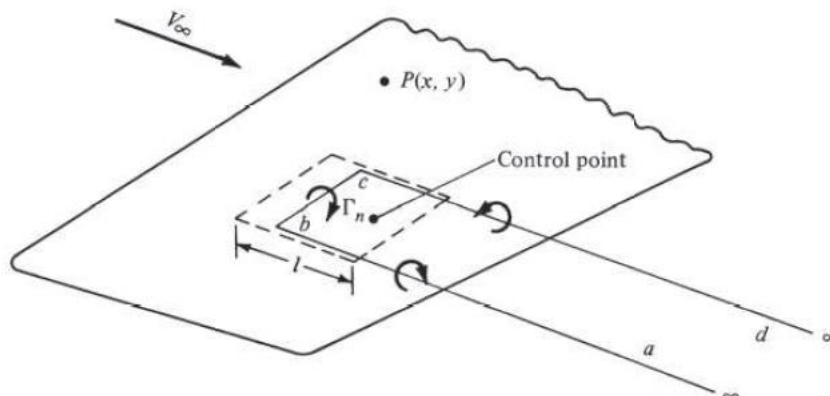


Vortex Lattice Method

- The Vortex Lattice Method (VLM) represents the wing as a surface on which a grid of horseshoe vortices (from lifting-line theory) is superimposed.
- The velocities induced by each horseshoe vortex at a specified control point are calculated using the Biot-Savart law.
- A summation is performed for all control points on the wing to produce a set of linear algebraic equations for the horseshoe vortex strengths that satisfy the boundary condition of no flow through the wing.
- The vortex strengths are related to the wing circulation and the pressure differential between the upper and lower wing surfaces.
- The pressure differentials are integrated to yield the total forces and moments.
- To solve the governing equation, the continuous distribution of bound vorticity over the wing surface is approximated by a finite number of discrete horseshoe vortices, as shown in Figure.



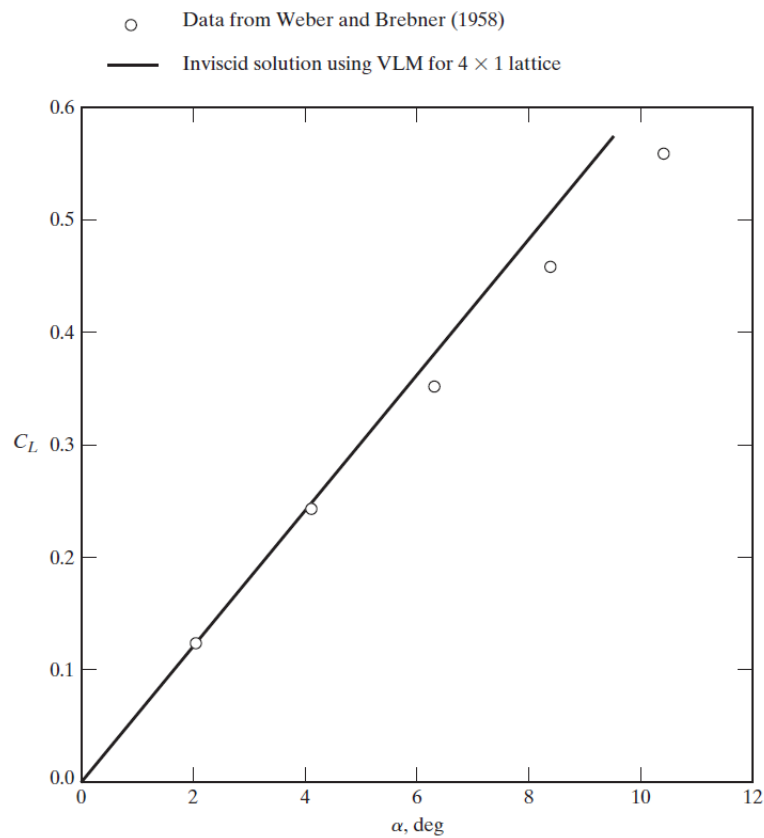
- Schematic of a single horseshoe vortex, which is part of a vortex system on the wing.



- The individual horseshoe vortices are placed in trapezoidal panels (also called *finite elements* or *lattices*).

- This procedure for obtaining a numerical solution to the flow is termed the vortex lattice method.
- The bound vortex typically coincides with the quarter-chord line of the panel (or element) and is, therefore, aligned with the local sweepback angle.
- In a rigorous theoretical analysis, the vortex lattice panels are located on the mean camber surface of the wing and, when the trailing vortices leave the wing, they follow a curved path.

Comparison of the theoretical and the experimental lift coefficients for the swept wing ($\Lambda = 45$ degree):



Reference:

1. Bertin, John J., and Russell M. Cummings. *Aerodynamics for engineers*. Cambridge University Press, 2021.
2. Anderson, J. (2011): *Fundamentals of Aerodynamics (SI units)*. McGraw hill.